MCQ Mock Test for 41107 Marine and Ocean Engineering

[Practice examination set generated from the uploaded mock test, homework notes, transcribed examples, and the updated formula sheet]

The scoring algorithm is based on **one best answer**.
This means:

- There is always only one correct answer.
- Students are only able to select one answer per question.
- Every correct answer corresponds to 1 point.
- Every incorrect answer corresponds to 0 points.

If not specifically stated, you may assume the following:

- Seawater density $\rho = 1025\,\mathrm{kg/m^3}$ and freshwater density $\rho = 1000\,\mathrm{kg/m^3}$.
- Kinematic viscosity $\nu = 1.0 \times 10^{-6}\,\mathrm{m^2/s}$.
- Deep-water conditions and long-crested waves where appropriate.
- Wave and response processes are Gaussian and peak values follow Rayleigh statistics where appropriate.

Mock exam

Part I — Flow around cylinders, VIV, and oscillatory-flow thinking

Question 1

FLOW AROUND AND FORCES ON CYLINDERS. Steady current around a free smooth circular cylinder. Consider a cylinder with $D = 1.0\,\mathrm{m}$ in a current $U_c = 0.25\,\mathrm{m/s}$. Take $\nu = 1.0 \times 10^{-6}\,\mathrm{m^2/s}$. Which flow regime is most appropriate?

A. Sub-critical

B. Critical / trans-critical onset

C. Super-critical

D. No vortex shedding because $Re < 1$

Question 2

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free circular cylinder. Consider the following three cases. Assume $C_D = 1.2$ in the sub-critical regime and $C_D = 0.9$ in the trans-critical regime. Which case gives the largest mean drag force per unit length?

A. $D = 0.3\,\mathrm{m},\ U_c = 0.4\,\mathrm{m/s}$

B. $D = 4.0\,\mathrm{m},\ U_c = 0.25\,\mathrm{m/s}$

C. $D = 0.8\,\mathrm{m},\ U_c = 0.6\,\mathrm{m/s}$

Question 3

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free cylinder. What is the pressure in excess of hydrostatic pressure at the stagnation point if $U_c = 1.6\,\mathrm{m/s}$ and $\rho = 1025\,\mathrm{kg/m^3}$?

A. 0 Pa

B. 820 Pa

C. 1310 Pa

D. 2620 Pa

Question 4

FLOW AROUND AND FORCES ON CYLINDERS. Plane oscillatory flow around a free smooth circular cylinder is described by $U(t) = U_m \sin(\omega t)$ with $U_m = 0.9\,\mathrm{m/s}$, $T = 6.0\,\mathrm{s}$ and $D = 0.60\,\mathrm{m}$. What is the Keulegan–Carpenter number?

A. $KC = 6.0$

B. $KC = 9.0$

C. $KC = 12.0$

D. $KC = 15.0$

Question 5

VORTEX INDUCED VIBRATIONS. Plane oscillatory flow around a free, smooth, flexibly-mounted circular cylinder. The oscillatory flow is $U(t) = 1.8 \sin(\omega t)$ m/s, the diameter is $D = 0.40\,\mathrm{m}$, and the angular natural frequency is $\omega_n = 9.42\,\mathrm{rad/s}$. What is the reduced velocity V_r?

A. $V_r = 2.0$

B. $V_r = 3.0$

C. $V_r = 4.5$

D. $V_r = 6.0$

Question 6

FLOW AROUND AND FORCES ON CYLINDERS. Plane oscillatory flow around a free smooth circular cylinder. For $U_m = 0.8\,\mathrm{m/s}$, $D = 0.50\,\mathrm{m}$ and $T = 4.0\,\mathrm{s}$, assume $C_D = 1.0$ and $C_M = 1.8$. What is the ratio $F_{I,\max}/F_{D,\max}$ approximately?

A. 0.28

B. 1.39

C. 2.78

D. 5.56

Question 7

FLOW AROUND AND FORCES ON CYLINDERS. Plane oscillatory flow around a free smooth circular cylinder. The normalized fundamental lift frequency is $N_L = 2$. Which name is given to this flow regime?

A. Single-pair regime

B. Double-pair regime

C. Three-pair regime

D. No-separation regime

Question 8

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free circular cylinder. If the vortex shedding frequency is $f_v = 0.24\,\text{Hz}$, what is the frequency of the oscillating drag force?

A. $0.12\,\text{Hz}$

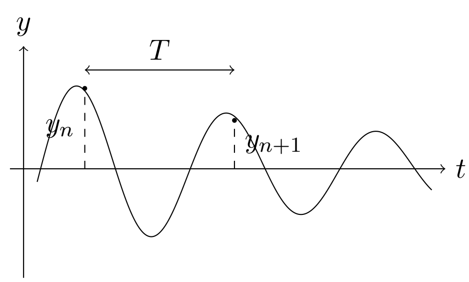
B. $0.24\,\text{Hz}$

C. $0.48\,\text{Hz}$

D. $0.96\,\text{Hz}$

Question 9

VORTEX INDUCED VIBRATIONS. The amplitude response in a free-decay test is shown in the figure. If $y_n = 1.5$, $y_{n+1} = 0.9$ and $T = 0.8\,\text{s}$, what are the damping ratio ζ and the undamped natural angular frequency ω_n approximately?



A. $\zeta = 0.081,\ \omega_n = 7.88\,\text{rad/s}$

B. $\zeta = 0.081,\ \omega_n = 6.28\,\text{rad/s}$

C. $\zeta = 0.163,\ \omega_n = 7.88\,\text{rad/s}$

D. $\zeta = 0.163,\ \omega_n = 6.28\,\text{rad/s}$

Question 10

VORTEX INDUCED VIBRATIONS. Calculate the undamped natural frequency of a flexibly mounted circular cylinder in water. Take $D = 0.08\,\text{m}$, structural mass per unit length $m = 4.0\,\text{kg/m}$, spring constant per unit length $k = 900\,\text{N/m}$, and assume $A/D < 0.8$ so that $c_m = 1$ for the added mass estimate. What is f_n approximately?

A. 0.79 Hz

B. 1.58 Hz

C. 2.25 Hz

D. 3.16 Hz

Question 11

DIFFRACTION / LARGE BODIES. For wave diffraction around a cylinder, when is diffraction expected to be important?

A. When $D/L \gtrsim 0.2$

B. When $D/L \lesssim 0.02$

C. Only when $KC > 20$

D. Only when $Re < 1000$

Question 12

FLOW AROUND AND FORCES ON CYLINDERS. For a smooth circular cylinder in the sub-critical regime, why is C_D almost constant over a wide range of Re?

A. Because separation occurs early and at nearly the same angular position, so the wake width stays similar.

B. Because friction drag dominates completely and pressure drag is negligible.

C. Because the boundary layer is turbulent from the stagnation point.

D. Because vortex shedding stops in the sub-critical regime.

Part II — Ship motion, hydrostatics, resistance, spectra, and strength

Question 13

SHIP MOTION DYNAMICS. Transfer functions of a ship are available. At a given frequency, the real and imaginary parts of the pitch RAO are $(\mathrm{Re}, \mathrm{Im}) = (-4.0 \times 10^{-2}, 3.0 \times 10^{-2})$ rad/m. What is the phase angle ε of the complex RAO, expressed in degrees in the interval $[0, 360)$?

A. 36.9°

B. 143.1°

C. 216.9°

D. 323.1°

Question 14

FORWARD SPEED / ENCOUNTERED WAVES. A ship travels at 18 knots in head seas. The true wave frequency is $f = 0.10$ Hz. What is the encountered period T_e approximately?

A. 6.3 s

B. 7.2 s

C. 8.5 s

D. 10.0 s

Question 15

SHIP MOTION DYNAMICS. A vertical circular cylinder floats with its axis vertical and undergoes free oscillations in pure heave. Damping is neglected. The draught measured from the bottom of the cylinder is $h = 0.80\,\text{m}$. What is the natural angular frequency approximately?

A. 2.5 rad/s

B. 3.5 rad/s

C. 4.5 rad/s

D. 5.5 rad/s

Question 16

SPECTRAL ANALYSIS. At a frequency ω_a, the wave energy spectral density is $S(\omega_a) = 1.8\,\text{m}^2\text{s/rad}$ and the corresponding pitch response spectral density is $R(\omega_a) = 0.008\,\text{rad}^2\text{s/rad}$. What is the magnitude of the pitch transfer function $|\Phi(\omega_a)|$ approximately?

A. 0.0667 rad/m (3.8 deg/m)

B. 0.0383 rad/m (2.2 deg/m)

C. 0.0067 rad/m (0.38 deg/m)

D. 0.1067 rad/m (6.1 deg/m)

Question 17

HYDROSTATIC COEFFICIENTS. A ship in saltwater has $L = 120\,\text{m}$, $B = 20\,\text{m}$, $T = 8\,\text{m}$ and displacement mass $\Delta = 15000\,\text{t}$. Taking $\rho = 1.025\,\text{t/m}^3$, what is the block coefficient C_B approximately?

A. 0.56

B. 0.68

C. 0.76

D. 0.91

Question 18

SHIP HYDROSTATICS. A rectangular, box-shaped barge has beam $B = 8.0\,\text{m}$ and floats at uniform draught T. What must the draught be for the transverse metacentre to lie in the waterplane?

A. $T = 2.5\,\text{m}$

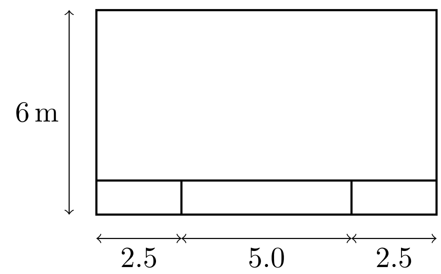
B. $T = 3.3\,\text{m}$

C. $T = 4.0\,\text{m}$

D. $T = 4.6\,\mathrm{m}$

Question 19

STRENGTH OF SHIPS. The thin-walled barge cross section shown below has uniform plate thickness $t = 10\,\mathrm{mm}$. What is the vertical distance z_n from the keel to the neutral axis?



A. $z_n = 2.32\,\mathrm{m}$

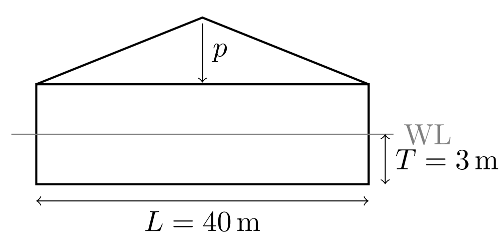
B. $z_n = 2.43\,\mathrm{m}$

C. $z_n = 2.57\,\mathrm{m}$

D. $z_n = 3.00\,\mathrm{m}$

Question 20

STRENGTH OF SHIPS. The barge below is loaded with gravel. The gravel load is triangular along the whole length with maximum intensity p at midship. The barge's own weight is uniformly distributed with intensity $0.2p$. The barge length is $40\,\mathrm{m}$, breadth is $6\,\mathrm{m}$, and it floats at draught $T = 3\,\mathrm{m}$ in freshwater. Approximately what value has p?



A. $p = 18\,\mathrm{tons/m}$

B. $p = 26\,\mathrm{tons/m}$

C. $p = 32\,\mathrm{tons/m}$

D. $p = 40\,\mathrm{tons/m}$

Question 21

SHIP RESISTANCE. A ship of length $180\,\mathrm{m}$ is represented by a model $4.0\,\mathrm{m}$ long for resistance estimation. If performance is compared at corresponding speeds, what is the ratio R_s/R_m approximately?

A. 9.1×10^3

B. 9.1×10^4

C. 9.1×10^5

D. 4.5×10^4

Question 22

SHIP RESISTANCE / MODEL TESTING. In towing-tank resistance estimation for ships with free-surface effects, what similarity must be preserved in practice?

A. Identical Froude numbers of model and ship

B. Identical Reynolds numbers of model and ship

C. Identical Froude and Reynolds numbers simultaneously

D. Identical Strouhal numbers only

Question 23

WAVE STATISTICS / SPECTRAL MOMENTS. A wave system has $H_s = 4.0\,\mathrm{m}$ and first spectral moment $m_1 = 0.80\,\mathrm{m}^2\,\mathrm{rad/s}$. What is a representative mean period T_1 approximately?

A. 6.3 s

B. 7.9 s

C. 9.4 s

D. 12.6 s

Question 24

WAVE STATISTICS. For an irregular wave record, the variance of the surface elevation is $m_0 = 1.44\,\mathrm{m}^2$. What is the probability that a wave crest exceeds the level $h_1 = 1.2\,\mathrm{m}$?

A. 13.5%

B. 36.8%

C. 60.7%

D. 77.9%